

Gyroscope Sensor

Objective:

To read and analyze angular velocity values along the X, Y, and Z axes using the built-in LSM9DS1 gyroscope sensor on the Arduino Nano 33 BLE.

Procedure:

1. Opened Arduino IDE.
2. Installed the Arduino_LSM9DS1 library.
3. Opened the SimpleGyroscope example from:
4. File → Examples → Arduino_LSM9DS1 → SimpleGyroscope
5. Uploaded the program to the Arduino Nano 33 BLE board.
6. Opened the Serial Monitor at 9600 baud rate.
7. Rotated and tilted the board in different directions.
8. Observed the gyroscope values displayed in the Serial Monitor.

Working Principle:

The gyroscope measures rotational movement (angular velocity) around three axes:

- X-axis → Rotation around X-axis
- Y-axis → Rotation around Y-axis
- Z-axis → Rotation around Z-axis

The values are measured in:

Degrees per second ($^{\circ}$ /s)

When the board is stationary:

$X \approx 0$

$Y \approx 0$

$Z \approx 0$

When the board is rotated, the values increase according to the speed and direction of rotation

Observations:

- Values changed when the board was rotated.
- Faster rotation produced larger values.
- Different axes responded depending on the direction of rotation.
- Small values were observed even when the board was stationary due to sensor noise.

When the board is stationary:

0.98	1.40	2.08
0.98	1.34	2.01
1.04	1.59	2.01
1.04	1.46	1.95

When the board is rotated:

95.46	11.29	62.74
76.35	19.47	46.81
58.53	26.25	45.65
52.80	28.08	54.26
68.91	29.11	84.05
92.04	24.29	113.89
111.21	10.62	139.10
125.55	-2.01	163.33
141.97	-11.23	180.18
152.77	-9.58	186.95
138.67	19.53	189.88